

Truth Tables: Negation, Conjunction & Disjunction

Logic & Reasoning Worksheet · Grade 9–11

Name: _____

Date: _____

Learning Objectives

- Construct and interpret truth tables for negation, conjunction, and disjunction
- Determine the truth value of compound statements using logical operations
- Combine negation with conjunction and disjunction to evaluate complex compound statements

Problems

1. Complete the truth table for the negation of P.

p	$\neg p$
T	
F	

2. Complete the truth table for the conjunction of P and Q.

p	q	$p \wedge q$
T	T	
T	F	
F	T	
F	F	

3. Complete the truth table for the disjunction of P and Q.

p	q	$p \vee q$
T	T	
T	F	



p	q	$p \vee q$
F	T	
F	F	

4. Let p = '10 is greater than 4' (True) and q = '3 is less than 5' (True). Find the truth value of the compound statement: p AND q.

5. Let p = '10 is greater than 4' (True) and q = '3 is less than 5' (True). Find the truth value of: NOT p AND q.

6. Let p = '10 is greater than 4' (True) and q = '3 is less than 5' (True). Find the truth value of: p OR NOT q.

7. Let p = '10 is greater than 4' (True) and q = '3 is less than 5' (True). Find the truth value of: NOT p OR NOT q.

8. Complete the truth table that includes both the conjunction and the negation of the conjunction.

p	q	$p \wedge q$	$\neg(p \wedge q)$
T	T		
T	F		
F	T		
F	F		

9. Complete the full truth table showing negations of each variable alongside their conjunction and disjunction.

p	q	$\neg p$	$\neg q$	$p \wedge q$	$p \vee q$
T	T				
T	F				



p	q	$\neg p$	$\neg q$	$p \wedge q$	$p \vee q$
F	T				
F	F				

10. Complete the three-variable truth table for the compound statement: p AND q AND r, along with NOT p and the disjunction of NOT p and r.

p	q	r	$\neg p$	$p \wedge q$	$p \wedge q \wedge r$	$\neg p \vee r$
T	T	T				
T	T	F				
T	F	T				
T	F	F				
F	T	T				
F	T	F				
F	F	T				
F	F	F				



Truth Tables: Negation, Conjunction & Disjunction — Answer Key

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Answer Key

1. Answer: See completed table

p	$\neg p$
T	F
F	T

- Negation reverses the truth value of a statement.
- If p is True, then $\neg p$ is False; if p is False, then $\neg p$ is True.

2. Answer: See completed table

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

- Conjunction (AND) is only TRUE when BOTH p and q are TRUE.
- All other combinations (TF, FT, FF) produce a FALSE result.

3. Answer: See completed table

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

- Disjunction (OR) is only FALSE when BOTH p and q are FALSE.
- All other combinations (TT, TF, FT) produce a TRUE result.

4. Answer: True



- Using the conjunction truth table, $T \wedge T = \text{TRUE}$.

5. Answer: False

- p is TRUE, so $\neg p$ is FALSE. q is TRUE.
- Using the conjunction truth table, $F \wedge T = \text{FALSE}$.

6. Answer: True

- p is TRUE. q is TRUE, so $\neg q$ is FALSE.
- Using the disjunction truth table, $T \vee F = \text{TRUE}$.

7. Answer: False

- p is TRUE so $\neg p$ is FALSE. q is TRUE so $\neg q$ is FALSE.
- Using the disjunction truth table, $F \vee F = \text{FALSE}$.

8. Answer: See completed table

p	q	$p \wedge q$	$\neg(p \wedge q)$
T	T	T	F
T	F	F	T
F	T	F	T
F	F	F	T

- First fill in the $p \wedge q$ column using the conjunction rule: only T when both are T.
- Then negate each value in the $p \wedge q$ column to get $\neg(p \wedge q)$.

9. Answer: See completed table

p	q	$\neg p$	$\neg q$	$p \wedge q$	$p \vee q$
T	T	F	F	T	T
T	F	F	T	F	T
F	T	T	F	F	T
F	F	T	T	F	F

- Fill $\neg p$ and $\neg q$ by reversing p and q respectively.
- Fill $p \wedge q$: only T when both p and q are T.
- Fill $p \vee q$: only F when both p and q are F.

10. Answer: See completed table



p	q	r	$\neg p$	$p \wedge q$	$p \wedge q \wedge r$	$\neg p \vee r$
T	T	T	F	T	T	T
T	T	F	F	T	F	F
T	F	T	F	F	F	T
T	F	F	F	F	F	F
F	T	T	T	F	F	T
F	T	F	T	F	F	T
F	F	T	T	F	F	T
F	F	F	T	F	F	T

- List all 8 combinations of T/F for p, q, r.
- $\neg p$: reverse p's value in each row.
- $p \wedge q$: TRUE only when both p and q are TRUE.
- $p \wedge q \wedge r$: TRUE only when $p \wedge q$ is TRUE AND r is TRUE (only row 1).
- $\neg p \vee r$: FALSE only when both $\neg p$ is FALSE (p is TRUE) and r is FALSE.

